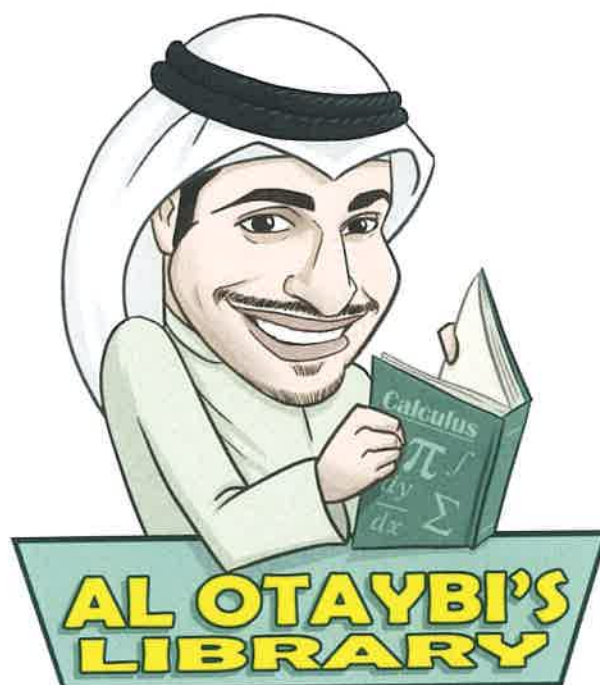


Kuwait University

2019/2020

Spring course



Calculus C

مراجعة

اختبارات الفيرست

(Oct 2019 – Mar 2019 – Oct 2018
June 2018 “version A” – June 2018 “version B”)

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1. (5+5+5+5 pts) Select the correct answer. No justification is required.

I. The value of $\lim_{(x,y) \rightarrow (0,\pi/4)} (x+y) \sin(x-2y)$ is:

- (A) $-\frac{\pi}{2}$ (B) 0 (C) $\frac{\pi}{4} - 1$ (D) $\frac{\pi}{4}$ (E) $-\frac{\pi}{4}$

II. If $\sin(y+z) + \ln x = 3 - z - y$, then $\frac{\partial z}{\partial y} =$

- (A) $-1 - \cos(y+z)$ (B) $-\cos(y+z)$ (C) -1
(D) $1 + \cos(y+z)$ (E) $\cos(y+z)$

III. The gradient of a differentiable function $f(x, y)$ is:

- (A) $f_x dx + f_y dy$ (B) $f_x \mathbf{i} + f_y \mathbf{j}$ (C) $f_x + f_y$
(D) $f_x u_1 + f_y u_2$ (E) $\mathbf{i} + \mathbf{j}$

IV. If $f(x, y, z) = \sinh^3(xyz)$, then $\frac{\partial f}{\partial x} =$

- (A) $3yz \sinh^2(xyz)$ (B) $3yz \cosh^2(xyz)$ (C) $yz \cosh^3(xyz)$
(D) $3yz \sinh^2(xyz) \cosh(xyz)$ (E) $3 \sinh^2(xyz) \cosh(xy^2 z^2)$

2. (10 pts) Find an equation for the tangent plane to the surface $z = \frac{x}{y^2}$ at $(3, 2, 3/4)$.

3. (10 pts) Find the limit, if it exists, or show that the limit does not exist:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \tan(xy)}{x^2 + 5y^2}$$

4. (10 pts) Find and sketch the domain of the function:

$$f(x, y) = \arcsin(x^2 + y^2 - 1)$$

5. (15 pts) Let $f(x, y) = ye^{-xy}$. Find all unit vectors $\mathbf{u}$, such that $D_{\mathbf{u}}f(0, 1) = 1$.

6. (15 pts) Show that $w = \sqrt{x^2 + y^2 + z^2}$ is a solution of the differential equation:

$$w_{xx} + w_{yy} + w_{zz} = \frac{2}{w}$$

7. (20 pts) Find the critical points of the function and classify them as local max/min or saddle points:

$$f(x, y) = x^3 + y^2 - 6xy + 6x + 3y - 2$$

The value of $\lim_{(x,y) \rightarrow (0, \frac{\pi}{4})} (x+y) \sin(x-2y)$ is :

Oct 2019
Question #1 I

- (A) $-\frac{\pi}{2}$ (B) 0 (C) $\frac{\pi}{4} - 1$ (D) $\frac{\pi}{4}$ (E) $-\frac{\pi}{4}$

$$\lim_{(x,y) \rightarrow (0, \frac{\pi}{4})} (x+y) \sin(x-2y)$$

$$= (0 + \frac{\pi}{4}) \sin(0 - 2(\frac{\pi}{4}))$$

$$= \frac{\pi}{4} \sin(-\frac{\pi}{2})$$

$$= -\frac{\pi}{4} \sin(\frac{\pi}{2}) = -\frac{\pi}{4} (1) = \boxed{-\frac{\pi}{4}} \text{ (E) } \checkmark$$

If $\sin(y+z) + \ln x = 3 - z - y$, then $\frac{\partial z}{\partial y} =$

Oct 2019
Question #1 II

- (A) $-1 - \cos(y+z)$ (B) $-\cos(y+z)$ (C) -1
(D) $1 + \cos(y+z)$ (E) $\cos(y+z)$

Let $F(x,y,z) = \sin(y+z) + \ln x - 3 + z + y$

$$\frac{\partial z}{\partial y} = \frac{-F_y}{F_z} ?$$

$$F_y = \cos(y+z) + 1$$

$$F_z = \cos(y+z) + 1$$

$$\Rightarrow \therefore \frac{\partial z}{\partial y} = \frac{-F_y}{F_z} = \frac{-[\cos(y+z) + 1]}{\cos(y+z) + 1} = -1$$

(C)
 ✓

The gradient of a differentiable function $f(x, y)$ is :

(A) $f_x dx + f_y dy$

(B) $f_x i + f_y j$

(C) $f_x + f_y$

(D) $f_x u_1 + f_y u_2$

(B) $i + j$

Oct 2019
Question 1 III

f_x

f_y



$\therefore$ the gradient vector

$$= \nabla f_x, f_y$$

$$= f_x i + f_y j \quad (B) \checkmark$$

If $f(x, y, z) = \sinh^3(xyz)$, then $\frac{\partial f}{\partial x} =$

(A) $3yz \sinh^2(xyz)$

(B) $3yz \cosh^2(xyz)$

(C) $yz \cosh^3(xyz)$

(D) $3yz \sinh^2(xyz) \cosh(xyz)$

(E) $3 \sinh^2(xyz) \cosh(xy^2 z^2)$

Oct 2019
Question 1 IV

$$f(x, y, z) = \sinh^3(xyz)$$

$$\frac{\partial f}{\partial x} = 3 \sinh^2(xyz) \cosh(xyz) \quad (1) yz$$

$$= 3 yz \sinh^2(xyz) \cosh(xyz)$$

(D) $\checkmark$

Find an equation for the tangent plane to the surface

$$z = \frac{x}{y^2} \text{ at } \left(3, 2, \frac{3}{4}\right)$$

Oct 2019
Question 2

$$f(x, y, z) = z - x y^{-2} \quad P\left(3, 2, \frac{3}{4}\right)$$

$$f_x = 0 - (1)y^{-2} = -\frac{1}{y^2} \rightarrow f_x\left(3, 2, \frac{3}{4}\right) = -\frac{1}{4}$$

$$f_y = 0 - x(-2)y^{-3} = \frac{2x}{y^3} \rightarrow f_y\left(3, 2, \frac{3}{4}\right) = \frac{6}{8} = \frac{3}{4}$$

$$f_z = 1 - 0 = 1 \rightarrow f_z\left(3, 2, \frac{3}{4}\right) = 1$$

$\therefore$ Equation of the tangent plane :

$$f_x(x - x_0) + f_y(y - y_0) + f_z(z - z_0) = 0$$

$$-\frac{1}{4}(x - 3) + \frac{3}{4}(y - 2) + (1)\left(z - \frac{3}{4}\right) = 0$$

sol #2

$$z = x y^{-2}$$

$$f_x = y^{-2} = \frac{1}{y^2} \rightarrow f_x\left(3, 2, \frac{3}{4}\right) = \frac{1}{4}$$

$$f_y = -2xy^{-3} = -\frac{2x}{y^3} \rightarrow f_y\left(3, 2, \frac{3}{4}\right) = -\frac{6}{8} = -\frac{3}{4}$$

$$\text{Eq. of T.P : } z - z_0 = f_x(x - x_0) + f_y(y - y_0)$$

$$z - \frac{3}{4} = \frac{1}{4}(x - 3) + \left(-\frac{3}{4}\right)(y - 2)$$

Find the limit, if it exist, or show that the limit does not exist :

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \tan(xy)}{x^2 + 5y^2} \quad \frac{0}{0}$$

Oct 2019

Question #3

$$0 \leq x^2 \leq x^2 + 5y^2$$

$$\div (x^2 + 5y^2)$$

$$0 \leq \frac{x^2}{x^2 + 5y^2} \leq 1$$

$$\times |\tan(xy)|$$

$$0 \leq \frac{x^2}{x^2 + 5y^2} |\tan(xy)| \leq |\tan(xy)|$$

$$\lim_{(x,y) \rightarrow (0,0)} 0 = 0$$

$$\lim_{(x,y) \rightarrow (0,0)} |\tan(xy)| = 0$$

∴ By Sandwich Theorem

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \tan(xy)}{x^2 + 5y^2} = 0$$

Find and sketch the domain of the function

Oct 2019

Question #4

$$f(x, y) = \arcsin(x^2 + y^2 - 1)$$

$$D_f = \{(x, y) \in \mathbb{R}^2 : -1 \leq x^2 + y^2 - 1 \leq 1\}$$

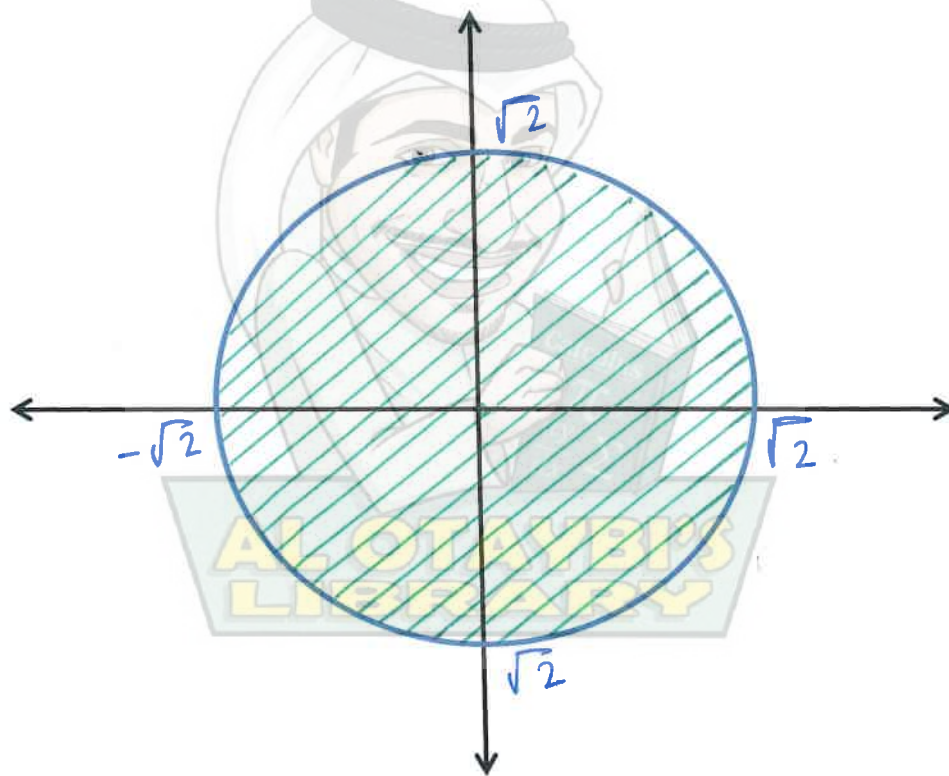
$$-1 \leq x^2 + y^2 - 1 \leq 1$$

$$0 \leq x^2 + y^2 \leq 2$$

$\mathbb{R}$

$$x^2 + y^2 \leq 2$$

$$\begin{array}{l} (0,0) \\ 0 \leq 2 \\ \checkmark \end{array}$$



Let $f(x, y) = ye^{-xy}$. Find all unit vectors u ,
such that $D_u f(0, 1) = 1$

Oct 2019
Question 5

$$f(x, y) = ye^{-xy} \quad P(0, 1)$$

$$f_x = -y^2 e^{-xy} \longrightarrow f_x(0, 1) = -(1)e^0 = -1$$

$$f_y = e^{-xy} - xye^{-xy} \longrightarrow f_y(0, 1) = 1 - 0 = 1$$

$$\Rightarrow \therefore \nabla f(0, 1) = \langle -1, 1 \rangle$$

$$\text{Let } \vec{u} = \langle u_1, u_2 \rangle$$

$$D_u f(0, 1) = \nabla f(0, 1) \cdot \vec{u} = 1$$

$$= \langle -1, 1 \rangle \cdot \langle u_1, u_2 \rangle = 1$$

$$-u_1 + u_2 = 1 \longrightarrow \boxed{u_2 = 1 + u_1} \longrightarrow \textcircled{1}$$

$$\|u\| = \sqrt{u_1^2 + u_2^2} = 1$$

$$u_1^2 + u_2^2 = 1 \longrightarrow \textcircled{2}$$

② in ①

$$u_1^2 + (1 + u_1)^2 = 1$$

$$u_1^2 + 1 + 2u_1 + u_1^2 = 1$$

$$2u_1^2 + 2u_1 = 0$$

$$2u_1(u_1 + 1) = 0$$

$$u_1 = 0 \xrightarrow{\text{in } \textcircled{1}} u_2 = 1$$

$$u_1 = -1 \xrightarrow{\text{in } \textcircled{1}} u_2 = 0$$

$$\Rightarrow \therefore u = \langle 0, 1 \rangle, u = \langle -1, 0 \rangle$$

Show that $w = \sqrt{x^2 + y^2 + z^2}$ is a solution of the differential equation :

$$w_{xx} + w_{yy} + w_{zz} = \frac{2}{w}$$

Oct 2019
Question 6

$$w_x = \frac{2x}{2\sqrt{x^2+y^2+z^2}} = \frac{x}{\sqrt{x^2+y^2+z^2}}$$

$$w_{xx} = \frac{\sqrt{x^2+y^2+z^2} (1) - x \cdot \frac{2x}{2\sqrt{x^2+y^2+z^2}}}{x^2+y^2+z^2}$$

$$= \frac{\sqrt{x^2+y^2+z^2} - \frac{x^2}{\sqrt{x^2+y^2+z^2}}}{x^2+y^2+z^2} = \frac{y^2+z^2}{\sqrt{x^2+y^2+z^2} (x^2+y^2+z^2)}$$

$$w_y = \frac{2y}{2\sqrt{x^2+y^2+z^2}} = \frac{y}{\sqrt{x^2+y^2+z^2}}$$

$$w_{yy} = \frac{\sqrt{x^2+y^2+z^2} (1) - y \cdot \frac{2y}{2\sqrt{x^2+y^2+z^2}}}{x^2+y^2+z^2}$$

$$= \frac{\sqrt{x^2+y^2+z^2} - \frac{y^2}{\sqrt{x^2+y^2+z^2}}}{x^2+y^2+z^2} = \frac{x^2+z^2}{\sqrt{x^2+y^2+z^2} (x^2+y^2+z^2)}$$

$$W_z = \frac{2z}{2\sqrt{x^2+y^2+z^2}} = \frac{z}{\sqrt{x^2+y^2+z^2}}$$

$$W_{zz} = \frac{\sqrt{x^2+y^2+z^2} (1) - z \frac{z}{\sqrt{x^2+y^2+z^2}}}{x^2+y^2+z^2}$$

$$= \frac{\frac{x^2+y^2+\cancel{z^2}-\cancel{z^2}}{\sqrt{x^2+y^2+z^2}}}{x^2+y^2+z^2} = \boxed{\frac{x^2+y^2}{\sqrt{x^2+y^2+z^2} (x^2+y^2+z^2)}}$$

$$L.H.S = W_{xx} + W_{yy} + W_{zz}$$

$$= \frac{y^2+z^2}{\sqrt{x^2+y^2+z^2} (x^2+y^2+z^2)} + \frac{x^2+z^2}{\sqrt{x^2+y^2+z^2} (x^2+y^2+z^2)} + \frac{x^2+y^2}{\sqrt{x^2+y^2+z^2} (x^2+y^2+z^2)}$$

$$= \frac{2x^2 + 2y^2 + 2z^2}{\sqrt{x^2+y^2+z^2} (x^2+y^2+z^2)}$$

$$= \frac{2(x^2+y^2+z^2)}{\sqrt{x^2+y^2+z^2} (x^2+y^2+z^2)}$$

$$= \frac{2}{\sqrt{x^2+y^2+z^2}} = R.H.S$$

Find the critical points of the function and classify them as local max/min or saddle points :

Oct 2019
Question 7

$$f(x, y) = x^3 + y^2 - 6xy + 6x + 3y - 2$$

$$f_x = 3x^2 - 6y + 6 \begin{cases} f_{xx} = 6x \\ f_{xy} = -6 \end{cases}$$

$$f_y = 2y - 6x + 3 \begin{cases} f_{yy} = 2 \\ f_{yx} = -6 \end{cases}$$

$$f_x = 0$$

$$3x^2 - 6y + 6 = 0$$

$$6y = 3x^2 + 6 \longrightarrow y = \frac{1}{2}x^2 + 1 \quad (1)$$

$$f_y = 0$$

$$2y - 6x + 3 = 0 \longrightarrow (2)$$

(1) in (2)

$$2\left(\frac{1}{2}x^2 + 1\right) - 6x + 3 = 0 \longrightarrow x^2 + 2 - 6x + 3 = 0$$

$$x^2 - 6x + 5 = 0$$

$$(x - 1)(x - 5) = 0 \begin{cases} x = 1 \xrightarrow{\text{in (1)}} y = \frac{1}{2} + 1 = \frac{3}{2} \\ x = 5 \xrightarrow{\text{in (1)}} y = \frac{25}{2} + 1 = \frac{27}{2} \end{cases}$$

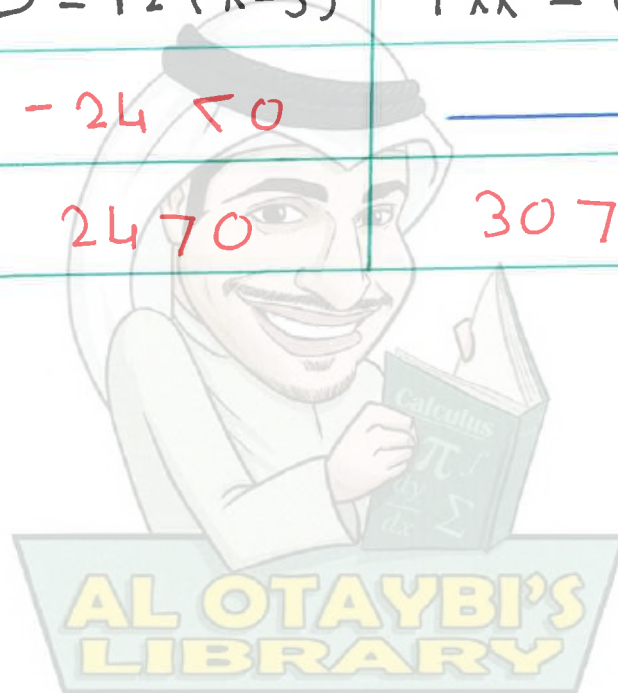
∴ The critical points are $(1, \frac{3}{2})$ and $(5, \frac{27}{2})$

$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx} f_{yy} - (f_{xy})^2$$

$$D = 6x(2) - (-6)^2 = 12x - 36$$

$$= 12(x - 3)$$

Critical Points	$D = 12(x - 3)$	$f_{xx} = 6x$	result
$(1, \frac{3}{2})$	$-24 < 0$	_____	Saddle point
$(5, \frac{27}{2})$	$24 > 0$	$30 > 0$	Local minimum



KUWAIT UNIVERSITY
Department of Mathematics
MATH 211: CALCULUS III

Spring 2018-19 • First Exam • 11 March 2019

1. [5 pts each] In each part, choose the correct answer.

i. $\lim_{(x,y) \rightarrow (0,0)} \frac{2x^4 + 3\cos(y^4)}{x^2 + y^2} = \underline{\hspace{2cm}}.$

- (a) 0 (b) 2 (c) 3 (d) it does not exist (e) 4

ii. The linearization of the function $f(x, y) = x \cos(xy) + \ln(x + y)$ at the point $(1, 0)$ is $\underline{\hspace{2cm}}.$

- (a) $2x + y + 1$ (b) $2x + y - 1$ (c) $2x - y + 3$
(d) $x + 2y - 1$ (e) $x + 2y + 1$

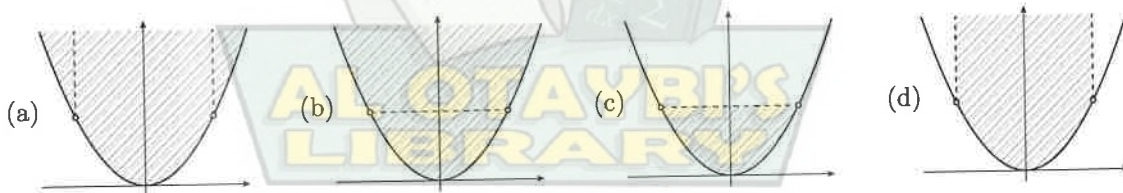
iii. If the equation $e^w = xy^w + 1$ defines w as a differentiable function of x and y , then $\frac{\partial w}{\partial x} = \underline{\hspace{2cm}}.$

- (a) $\frac{-y^w}{e^w - xy^{w-1} \ln y}$ (b) $\frac{e^w - y^w}{e^w - xy^w \ln y}$ (c) $y^w + xy^w \ln y$
(d) $\frac{y^w}{e^w - xy^w \ln y}$ (e) $\frac{y^w}{e^w - xy^w \ln y}$

iv. If $f(x, y) = \sqrt{1 + xy^3}$, then $f_{xx} = \underline{\hspace{2cm}}.$

- (a) $\frac{1}{4(1 + xy^3)^{3/2}}$ (b) $\frac{y^6}{4(1 + xy^3)^{3/2}}$ (c) $\frac{-y^6}{4(1 + xy^3)^{3/2}}$
(d) $\frac{3y^6}{4(1 + xy^3)^{3/2}}$ (e) $\frac{-3y^6}{4(1 + xy^3)^{3/2}}$

v. Which of the following sketches best describes the domain of the function $f(x, y) = \frac{\sqrt{y - x^2}}{4 - x^2}$?



2. [15 pts] Let $u = \sqrt{x^2 + y^2}$, where $x = r + s \cos t$ and $y = s + r \sin t$. Find $\frac{\partial u}{\partial r}$ and $\frac{\partial u}{\partial t}$ when $r = 2$, $s = 1$, and $t = 0$.

3. [15 pts] Find an equation for the tangent plane at the point $(1, -1, 1)$ to the surface

$$x^4 + y^4 - z^4 = x^2 y^2 z^2.$$

4. [15 pts] Find the absolute maximum and minimum values of $f(x, y) = x^2 + y^2 + 2x - 4y$ on the closed triangular region with vertices at $(0, 0)$, $(3, 0)$ and $(0, 3)$.

5. [15 pts] Find the point on the plane $x + 2y + 3z = 28$ that is closest to the origin.

6. [10+5 pts] Consider the function $f(x, y) = 5xe^y - x^2y$.

- (a) Find the rate of change of f at the point $(1, 0)$ in the direction of the vector $\mathbf{i} + 3\mathbf{j}$.
(b) Find the maximum value of the directional derivative of f at the point $(1, 0)$.

Mar 2019
Question #1 i

$$\lim_{(x,y) \rightarrow (0,0)} \frac{2x^4 + 3\cos(y^4)}{x^2 + y^2} =$$

- (A) 0 (B) 2 (C) 3 (D) it does not exist (E) 4

$$\lim_{(x,y) \rightarrow (0,0)} \frac{2x^4 + 3\cos(y^4)}{x^2 + y^2} = \frac{0 + 3(1)}{0} = \frac{3}{0} = \infty$$

DNE

(D) ✓

The linearization of the function $f(x, y) = x\cos(xy) + \ln(x + y)$ at the point $(1, 0)$ is :

- (a) $2x + y + 1$ (b) $2x + y - 1$ (c) $2x - y + 3$
(d) $x + 2y - 1$ (e) $x + 2y + 1$

Mar 2019
Question #1 ii

$$f(x, y) = x\cos(xy) + \ln(x + y) \quad P(1, 0)$$

$$x_0 = 1 \rightarrow \Delta x = x - 1$$

$$y_0 = 0 \rightarrow \Delta y = y$$

$$f(x_0, y_0) = (1)\cos(0) + \ln(1) = 1$$

$$f_x = \cos(xy) - xy \sin(xy) + \frac{1}{x+y}$$

$$f_x(x_0, y_0) = 1 - 0 + \frac{1}{1+0} = 2$$

$$f_y = -x^2 \sin(xy) + \frac{1}{x+y}$$

$$f_y(x_0, y_0) = 0 + \frac{1}{1} = 1$$

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0) \Delta x + f_y(x_0, y_0) \Delta y$$

$$= 1 + 2(x-1) + 1(y) = 1 + 2x - 2 + y = 2x + y - 1 \quad (b) \checkmark$$

If the equation $e^w = xy^w + 1$ defines w as

a differentiable function of x and y , then $\frac{\partial w}{\partial x} =$

(a) $\frac{-y^w}{e^w - xy^{w-1} \ln y}$

(b) $\frac{e^w - y^w}{e^w - xwy^{w-1}}$

(c) $y^w + xy^w \ln y$

(d) $\frac{y^w}{e^w - xwy^{w-1}}$

(e) $\frac{y^w}{e^w - xy^w \ln y}$

Mar 2019
Question 1 iii

Let $F = e^w - xy^w - 1$

$$\frac{\partial w}{\partial x} = -\frac{F_x}{F_w}$$

$$F_w = e^w - xy^w \cdot \ln y$$

$$F_x = 0 - (1)y^w = -y^w$$

$$\Rightarrow \frac{\partial w}{\partial x} = \frac{-F_x}{F_w} = \frac{y^w}{e^w - xy^w \ln y}$$

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LIBRARY

(e) ✓

If $f(x, y) = \sqrt{1 + xy^3}$ then $f_{xx} =$

(a) $\frac{1}{4(1 + xy^3)^{\frac{3}{2}}}$

(b) $\frac{y^6}{4(1 + xy^3)^{\frac{3}{2}}}$

(c) $\frac{-y^6}{4(1 + xy^3)^{\frac{3}{2}}}$

(d) $\frac{3y^6}{4(1 + xy^3)^{\frac{3}{2}}}$

(e) $\frac{-3y^6}{4(1 + xy^3)^{\frac{3}{2}}}$

Mar 2019
Question 1 iv

$$f(x, y) = \sqrt{1 + xy^3}$$

$$f_x = \frac{(1) y^3}{2 \sqrt{1 + xy^3}}$$

$$f_{xx} = \frac{0 - y^3 \cdot 2 \frac{(1) y^3}{2 \sqrt{1 + xy^3}}}{(2 \sqrt{1 + xy^3})^2}$$

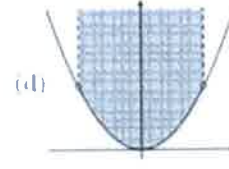
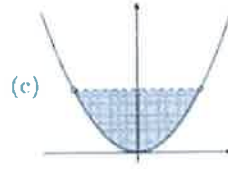
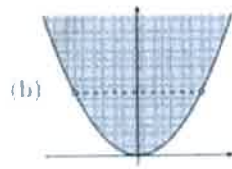
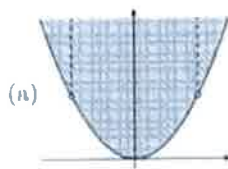
$$= \frac{-y^6}{4 (1 + xy^3)}$$

$$= \frac{-y^6}{4 (1 + xy^3)^{\frac{3}{2}}}$$

(c)

v. Which of the following sketches best describes the domain of the function $f(x, y) = \frac{\sqrt{y - x^2}}{4 - x^2}$?

Mar 2019
Question #1.v



$$f(x, y) = \frac{\sqrt{y - x^2}}{4 - x^2}$$

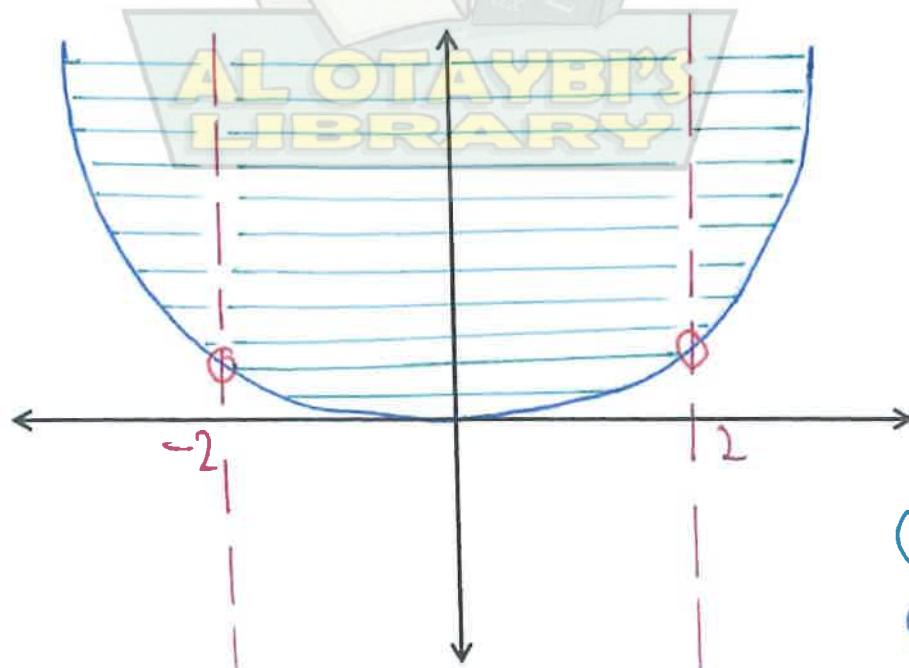
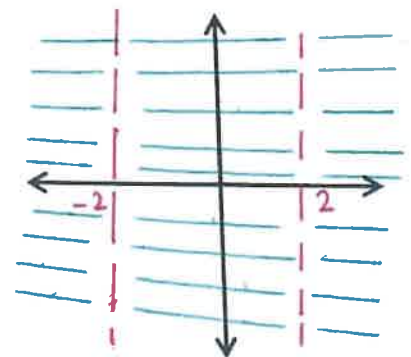
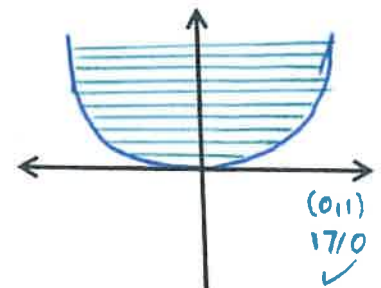
$$D_f = \{(x, y) \in \mathbb{R}^2 : y - x^2 \geq 0 \text{ and } 4 - x^2 \neq 0\}$$

$$y \geq x^2$$

$$4 - x^2 \neq 0$$

$$x^2 \neq 4$$

$$x \neq -2 \quad x \neq 2$$



(a) ✓

Let $u = \sqrt{x^2 + y^2}$ where $x = r + s \cos t$ and $y = s + r \sin t$

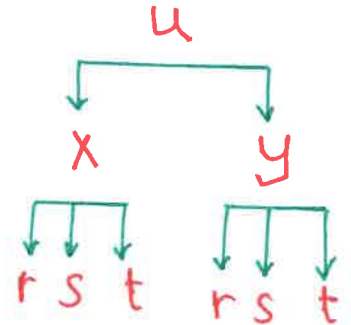
Find $\frac{\partial u}{\partial r}$ and $\frac{\partial u}{\partial t}$ when $r = 2, s = 1$ and $t = 0$.

Mar 2019
Question #2

$$\begin{aligned} r &= 2 \\ s &= 1 \\ t &= 0 \end{aligned}$$

$$x = 2 + (1)(1) = 3 \rightarrow x = 3$$

$$y = 1 + 0 = 1 \rightarrow y = 1$$



$u_r ? \quad u_t ?$

$$\begin{aligned} u_r &= u_x x_r + u_y y_r \\ &= \frac{2x}{2\sqrt{x^2 + y^2}} (1) + \frac{2y}{2\sqrt{x^2 + y^2}} \sin t \end{aligned}$$

at $r=2 \quad s=1 \quad t=0 \quad x=3 \quad y=1$

$$u_r = \frac{3}{\sqrt{9+1}} + 0 = \frac{3}{\sqrt{10}}$$

$$u_t = u_x x_t + u_y y_t$$

$$= \frac{2x}{2\sqrt{x^2 + y^2}} (-s \sin t) + \frac{2y}{2\sqrt{x^2 + y^2}} (r \cos t)$$

at $r=2 \quad s=1 \quad t=0 \quad x=3 \quad y=1$

$$u_t = 0 + \frac{1}{\sqrt{9+1}} (2(1)) = \frac{2}{\sqrt{10}}$$

Find an equation for the tangent plane to the point $(1, -1, 1)$
to the surface $x^4 + y^4 - z^4 = x^2 y^2 z^2$

Mar 2019
Question 3

$$F(x, y, z) = x^4 + y^4 - z^4 - x^2 y^2 z^2 \quad P(1, -1, 1)$$

$$F_x = 4x^3 - 2xy^2z^2 \rightarrow F_x(1, -1, 1) = 4 - 2 = 2$$

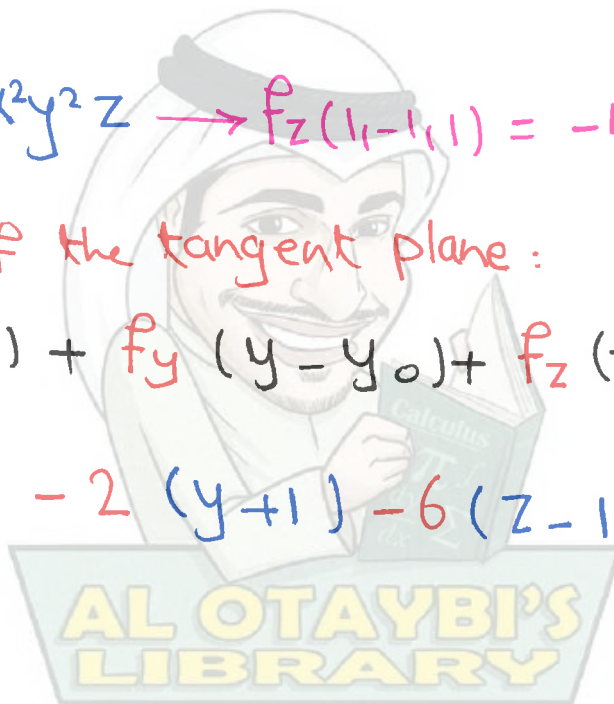
$$F_y = 4y^3 - 2x^2yz^2 \rightarrow F_y(1, -1, 1) = -4 + 2 = -2$$

$$F_z = -4z^3 - 2x^2y^2z \rightarrow F_z(1, -1, 1) = -4 - 2 = -6$$

$\therefore$ Equation of the tangent plane:

$$F_x(x - x_0) + F_y(y - y_0) + F_z(z - z_0) = 0$$

$$2(x - 1) - 2(y + 1) - 6(z - 1) = 0$$



Find the absolute maximum and minimum values of

Mar 2019
Question 4

$f(x, y) = x^2 + y^2 + 2x - 4y$ on the closed triangular region with vertices at $(0, 0)$, $(3, 0)$ and $(0, 3)$.

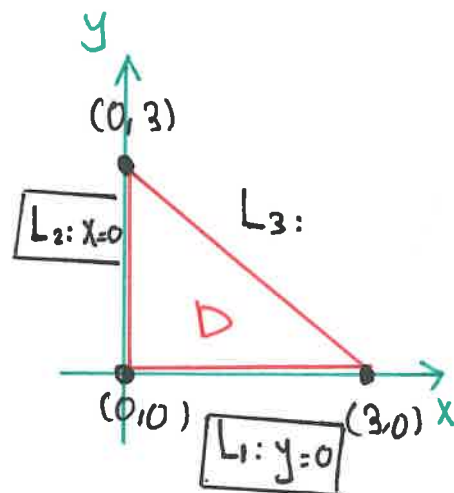
$$f_x = 2x + 2$$

$$f_x = 0 \longrightarrow 2x + 2 = 0 \longrightarrow x = -1$$

$$f_y = 2y - 4$$

$$f_y = 0 \longrightarrow 2y - 4 = 0 \longrightarrow y = 2$$

$$P(-1, 2) \notin D$$



along $L_1: y = 0$

$$f(x) = x^2 + 0 + 2x - 0$$

$$f'(x) = 2x + 2$$

$$\implies f'(x) = 0 \longrightarrow 2x + 2 = 0 \longrightarrow x = -1 \longrightarrow y = 1 - 2 = -1$$

$$P(-1, -1) \notin D$$

along $L_2: x = 0$

$$f(y) = 0 + y^2 + 0 - 4y$$

$$f'(y) = 2y - 4$$

$$\implies f'(y) = 0 \longrightarrow 2y - 4 = 0 \longrightarrow y = 2 \longrightarrow x = 4 - 8 = -4$$

$$P(-4, 2) \notin D$$

along $L_3: P_1(0, 3) \quad P_2(3, 0)$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 - 3}{3 - 0} = -\frac{3}{3} = -1$$

equation of L_3 :

$$y - y_0 = m(x - x_0)$$

$$y - 3 = -1(x - 0)$$

$$y = -x + 3$$

along $L_3: y = -x + 3$

$$\begin{aligned} F(x) &= x^2 + (-x+3)^2 + 2x - 4(-x+3) \\ &= x^2 + x^2 - 6x + 9 + 2x + 4x - 12 \\ &= 2x^2 - 3 \end{aligned}$$

$$F'(x) = 6x$$

$$\Rightarrow F'(x) = 0 \rightarrow 6x = 0 \rightarrow x = 0 \rightarrow y = -3$$

$$P(0, -3) \notin D$$

The intersection points $P(0,0)$ $P(0,3)$ $P(3,0)$

$$F(0,0) = 0 + 0 + 0 - 0 = 0$$

$$F(0,3) = 0 + 9 + 0 - 12 = -3 \rightarrow \text{abs. minimum}$$

$$F(3,0) = 9 + 0 + 6 - 0 = 15 \rightarrow \text{abs. maximum}$$

Find the points on the plane $x + 2y + 3z = 28$ that is closest to the origin.

Mar 2019
Question 5

$$L^2 = (x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2 \quad P(0,0,0) \quad g(x,y,z) = x + 2y + 3z - 28 = 0$$

$$L^2 = (x-0)^2 + (y-0)^2 + (z-0)^2$$

$$L^2 = x^2 + y^2 + z^2$$

$$\text{Let } f(x,y,z) = x^2 + y^2 + z^2 \quad g(x,y,z) = x + 2y + 3z - 28 = 0$$

$$\nabla f = \langle 2x, 2y, 2z \rangle \quad \nabla g = \langle 1, 2, 3 \rangle$$

$$\rightarrow \nabla f = \lambda \nabla g$$

$$\langle 2x, 2y, 2z \rangle = \lambda \langle 1, 2, 3 \rangle$$

$$\begin{array}{l|l|l} 2x = \lambda & 2y = 2\lambda & 2z = 3\lambda \\ (1) & (2) & (3) \end{array}$$

(1) in (2)

$$2y = 2(2x) \xrightarrow{\div 2} \boxed{y = 2x}$$

(1) in (3)

$$2z = 3(2x) \xrightarrow{\div 2} \boxed{z = 3x}$$

$$y = 2x \xrightarrow{\text{in } g} x + 2(2x) + 3(3x) - 28 = 0$$

$$z = 3x$$

$$x + 4x + 9x - 28 = 0$$

$$14x = 28 \rightarrow x = 2$$

$$y = 4 \quad z = 6 \rightarrow \boxed{P(2, 4, 6)}$$

$$f(2, 4, 6) = 4 + 16 + 36 = 56$$

$\therefore$ The point $P(2, 4, 6)$ is closest point to the origin

Consider the function $f(x, y) = 5xe^y - x^2y$

a) Find the rate of change of f at the point $(1, 0)$
in the direction of the vector $i + 3j$.

b) Find the maximum value of the directional
derivative of f at the point $(1, 0)$

Mar 2019

Question 6

$$f(x, y) = 5xe^y - x^2y \quad P(1, 0) \quad v = \langle 1, 3 \rangle$$

$$f_x = 5e^y - 2xy \longrightarrow f_x(1, 0) = 5e^0 - 0 = 5$$

$$f_y = 5xe^y - x^2 \longrightarrow f_y(1, 0) = 5e^0 - 1 = 4 \implies \nabla f(1, 0) = \langle 5, 4 \rangle$$

$$\vec{u}_v = \frac{\vec{v}}{\|\vec{v}\|} = \frac{\langle 1, 3 \rangle}{\sqrt{1+9}} = \frac{\langle 1, 3 \rangle}{\sqrt{10}} = \left\langle \frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}} \right\rangle$$

$$D_{\vec{u}} f(1, 0) =$$

$$= \langle 5, 4 \rangle \cdot \left\langle \frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}} \right\rangle =$$

$$= \frac{5}{\sqrt{10}} + \frac{12}{\sqrt{10}} = \frac{17}{\sqrt{10}}$$

$$b) [D_{\vec{u}} f(1, 0)]_{\max} = \|\nabla f\|$$

$$= \sqrt{25 + 16} = \sqrt{41}$$

Math 211 First Exam

14 October 2018

Questions 1–5: 5 points each, questions 6–10: 15 points each.

In questions 1–5, select the correct Answer. Use back of the previous page for your calculations.

1. Let $\mathbf{u}(t)$ and $\mathbf{v}(t)$ be differentiable vector-valued functions and $f(t)$ be a differentiable real-valued function. Then $\frac{d}{dt}[f(t)[\mathbf{u}(t) \times \mathbf{v}(t)]]$ equals _____.

(a) $f(t)[\mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)]$

(b) $f(t)[\mathbf{u}'(t) \times \mathbf{v}'(t)]$

(c) $f'(t)[\mathbf{u}'(t) \times \mathbf{v}'(t)]$

(d) $f'(t)[\mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)]$

(e) $f'(t)[\mathbf{u}(t) \times \mathbf{v}(t)] + f(t)[\mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)]$

2. The length of the curve with vector equation $\mathbf{r}(t) = \langle t^2, 2t, \ln t \rangle$, $1 \leq t \leq 3$ is _____.

(a) $12 + \ln 3$

(b) $8 + \ln 3$

(c) $2 + \ln 3$

(d) $4 + 2 \ln 3$

(e) 2

3. Given $f(x, y) = \frac{y^2 \sin x}{x^2 + y^4}$, which of the following is false?

(a) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ along the x -axis equals 0.

(b) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ along the y -axis equals 0.

(c) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ along the line $y = x$ equals 0.

(d) $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$.

(e) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist.

4. Which function satisfies the equation $u_{xx} + u_{yy} = 0$?

(a) $u(x, y) = x^2 + y^3$

(b) $u(x, y) = \sinh x \cosh y$

(c) $u(x, y) = \ln \sqrt{x^2 + y^2}$

(d) $u(x, y) = \sin x \cos y$

(e) $u(x, y) = x^3 y + 3x^2 y^2$

5. Let $F(x, y, z)$ be a differentiable function satisfying $F_x - F_y = 0$. Suppose that the equation $F(x, y, z) = 0$ defines z implicitly as a differentiable function of x and y , then _____.

(a) $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial y}$

(b) $\frac{\partial z}{\partial x} = -\frac{\partial z}{\partial y}$

(c) $\frac{\partial z}{\partial x} \frac{\partial z}{\partial y} = 1$

(d) $\frac{\partial z}{\partial x} \frac{\partial z}{\partial y} = -1$

(e) $\frac{\partial x}{\partial z} = \frac{\partial z}{\partial y}$

6. Let $f(x)$ be a function with continuous second derivative and let C be the graph $y = f(x)$.

(a) Write a vector function that represents the curve C .

(b) Prove that the curvature at any point on C is $\kappa(x) = \frac{|y''|}{[1 + (y')^2]^{3/2}}$.

7. Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + y^2}$ exists.

8. Let $f(x, y) = x \cos y - 2^x \sin y$.

(a) Show that f is differentiable on $\mathbb{R}^2$.

(b) Find an equation for the tangent plane to the surface $z = f(x, y)$ at the point $(1, \pi, -1)$.

(c) Use the linearization of f at $(1, \pi)$ to estimate $f(1.1, 3)$.

9. Let $f(x, y)$ be a function whose second-order partial derivatives are continuous throughout $\mathbb{R}^2$. If $z = f(x, y)$ where $x = st$, and $y = 2s + 3t$, find $\frac{\partial z}{\partial t}$ and $\frac{\partial^2 z}{\partial s \partial t}$ in terms of partial derivatives of f .

10. Find all points on the surface $x^2 - y^2 - z^2 = 4$ where the tangent plane is parallel to the plane $3x + 2y + z = 7$.

Given $f(x, y) = \frac{y^2 \sin x}{x^2 + y^4}$, which of the following is false?

- (a) $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$ along the x -axis equals 0
- (b) $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$ along the y -axis equals 0
- (c) $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$ along the line $y = x$ equals 0
- (d) $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$
- (e) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist

Oct 2018
Question #3

$$\lim_{(x,y) \rightarrow (0,0)} \frac{y^2 \sin x}{x^2 + y^4} \quad \frac{0}{0}$$

along $x=0$ (y -axis)

$$\lim_{y \rightarrow 0} \frac{0}{0 + y^4} = 0$$

along $x=y^2$

$$\lim_{y \rightarrow 0} \frac{y^2 \sin y^2}{(y^2)^2 + y^4} = \lim_{y \rightarrow 0} \frac{y^2 \sin y^2}{2y^4}$$

$$\lim_{y \rightarrow 0} \frac{\sin y^2}{2y^2} = \frac{1}{2}$$

$$\therefore \lim_{(x,y) \rightarrow (0,0)} \frac{y^2 \sin x}{x^2 + y^4} \text{ DNE (two path rule)}$$

Which function satisfies the equation $u_{xx} + u_{yy} = 0$?

- (a) $u(x, y) = x^2 + y^3$ (b) $u(x, y) = \sinh x \cosh y$ (c) $\ln \sqrt{x^2 + y^2}$
 (d) $u(x, y) = \sin x \cos x$ (e) $u(x, y) = x^3 y + 3x^2 y^2$

Oct 2018
Question 4

(a) $u(x, y) = x^2 + y^3$

$$\begin{aligned} u_x &= 2x & u_y &= 3y^2 \\ u_{xx} &= 2 & u_{yy} &= 6y \end{aligned} \Rightarrow u_{xx} + u_{yy} = 2 + 6y$$

(b) $u(x, y) = \sinh x \cosh y$

$$u_x = \cosh x \cosh y$$

$$u_{xx} = \sinh x \cosh y$$

$$u_y = \sinh x \sinh y$$

$$u_{yy} = \sinh x \cosh y$$

$$u_{xx} + u_{yy}$$

$$= \sinh x \cosh y + \sinh x \cosh y = 2 \sinh x \cosh y$$

(c) $u(x, y) = \ln(x^2 + y^2)^{\frac{1}{2}} = \frac{1}{2} \ln(x^2 + y^2)$

$$u_x = \frac{1}{2} \cdot \frac{2x}{x^2 + y^2}$$

$$u_{xx} = \frac{(x^2 + y^2)(1) - x \cdot (2x)}{(x^2 + y^2)^2} = \frac{x^2 + y^2 - 2x^2}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$u_y = \frac{1}{2} \cdot \frac{2y}{x^2 + y^2}$$

$$u_{yy} = \frac{(x^2 + y^2)(1) - y \cdot (2y)}{(x^2 + y^2)^2} = \frac{x^2 + y^2 - 2y^2}{(x^2 + y^2)^2} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$$

$$\Rightarrow u_{xx} + u_{yy}$$

$$= \frac{y^2 - x^2}{(x^2 + y^2)^2} + \frac{x^2 - y^2}{(x^2 + y^2)^2} = \boxed{0}$$

(c) ✓

Let $f(x, y, z)$ be a differentiable function satisfying $F_x - F_y = 0$.

Suppose that the equation $F(x, y, z) = 0$ defines z implicitly as a differentiable function of x and y , then :

(a) $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial y}$

(b) $\frac{\partial z}{\partial x} = -\frac{\partial z}{\partial y}$

(c) $\frac{\partial z}{\partial x} \frac{\partial z}{\partial y} = 1$

(d) $\frac{\partial z}{\partial x} \frac{\partial z}{\partial y} = -1$

(e) $\frac{\partial x}{\partial z} = \frac{\partial z}{\partial y}$

Oct 2018
Question 5

$$F_x - F_y = 0 \rightarrow \therefore F_x = F_y$$
$$\frac{\partial z}{\partial x} = \frac{-F_x}{F_z} = \frac{-F_y}{F_z}$$
$$\frac{\partial z}{\partial y} = \frac{-F_y}{F_z}$$
$$\therefore \frac{\partial z}{\partial x} = \frac{\partial z}{\partial y} \quad (a)$$

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Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + y^2}$ exists.

Oct 2018
Question #7

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + y^2} = \frac{0}{0}$$

$$0 < y^2 < x^4 + y^2$$

$$* \frac{x^2}{x^4 + y^2}$$

$$0 < \frac{x^2}{x^4 + y^2}$$

$$\lim_{(x,y) \rightarrow (0,0)} 0 = 0$$

$$\frac{x^2 y^2}{x^4 + y^2} < x^2$$

$$\lim_{(x,y) \rightarrow (0,0)} x^2 = 0$$

∴ By Sandwich theorem

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + y^2} = 0 \quad (\text{exists})$$

Let $f(x, y) = x \cos y - 2^x \sin y$

Oct 2018

Question 8

(a) Show that f is differentiable on $\mathbb{R}^2$.

(b) Find an equation for the tangent plane to the surface $z = f(x, y)$ at the point $(1, \pi, -1)$.

(c) Use the linearization of f at $(1, \pi)$ to estimate $f(1.1, 3)$.

a) $f(x, y) = x \cos y - 2^x \sin y$

$$f_x = \cos y - 2^x \ln 2 \sin y$$

$\Rightarrow \therefore f_x$ is exist and continuous on $\mathbb{R}^2 \rightarrow \textcircled{1}$

$$f_y = -x \sin y - 2^x \cos y$$

$\Rightarrow \therefore f_y$ is exist and continuous on $\mathbb{R}^2 \rightarrow \textcircled{2}$

From $\textcircled{1} \textcircled{2} \rightarrow \therefore f$ is differentiable on $\mathbb{R}^2$

b) $z = x \cos y - 2^x \sin y$ $P(1, \pi, -1)$

Let $f(x, y, z) = x \cos y - 2^x \sin y - z$

$$f_x = \cos y - 2^x \ln 2 \sin y \rightarrow f_x(1, \pi, -1) = -1 - 0 = -1$$

$$f_y = -x \sin y - 2^x \cos y \rightarrow f_y(1, \pi, -1) = 0 - 2^1(-1) = 2$$

$$f_z = -1 \rightarrow f_z(1, \pi, -1) = -1$$

$\therefore$ The Equation of the tangent plane :

$$f_x(x - x_0) + f_y(y - y_0) + f_z(z - z_0) = 0$$

$$-1(x - 1) + 2(y - \pi) + 2(z - (-1)) = 0$$

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$$(c) \quad f(x, y) = x \cos y - 2^x \sin y \quad P(1, \pi)$$

$$x_0 = 1$$

$$x_1 = x \rightarrow \Delta x = x - 1$$

$$y_0 = \pi$$

$$y_1 = y \rightarrow \Delta y = y - \pi$$

$$1) \quad f(x_0, y_0) = (1) \cos \pi - 0 = -1$$

$$2) \quad f_x(1, \pi) = -1 \quad f_y(1, \pi) = 2$$

$$\begin{aligned} L(x_1, y_1) &= f(x_0, y_0) + f_x(x_0, y_0) \Delta x + f_y(x_0, y_0) \Delta y \\ &= -1 + (-1)(x-1) + 2(y-\pi) \end{aligned}$$

$$\text{at } P(1.1, 3)$$

$$\Delta f(x, y) \approx -1 - (1.1 - 1) + 2(3 - \pi)$$

$$\approx -1 - 0.1 + 6 - 2\pi$$

$$\approx 4.9 - 2\pi \quad \rightarrow 2(3.14)$$

$$\approx 4.9 - 6.28$$

$$\approx -1.38$$

Let $f(x, y)$ be a function whose second-order partial derivatives are continuous throughout R^2 . If $z = f(x, y)$ where

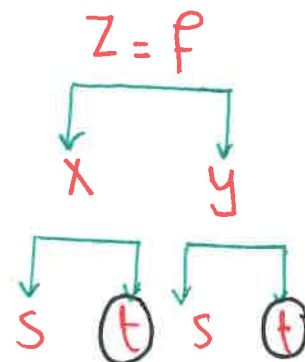
$$x = st \quad \text{and} \quad y = 2s + 3t, \quad \text{Find } \frac{\partial z}{\partial t} \quad \text{and} \quad \frac{\partial^2 z}{\partial s \partial t}$$

in terms of partial derivatives of f .

Oct 2018
Question #9

$$Z_t \quad ?? \quad Z_{ts} \quad ??$$

$$\begin{aligned} Z_t &= Z_x \cdot X_t + Z_y \cdot Y_t \\ &= Z_x (s) + Z_y (3) \end{aligned}$$

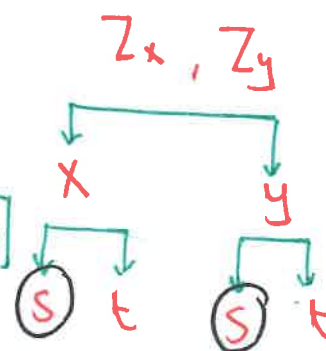


$$\begin{aligned} Z_{ts} &= \frac{\partial}{\partial s} [s Z_x + 3 Z_y] \\ &= (1) Z_x + s \frac{\partial}{\partial s} Z_x + 3 \frac{\partial}{\partial s} Z_y \\ &= Z_x + s [Z_{xx} X_s + Z_{xy} Y_s] + 3 [Z_{yx} X_s + Z_{yy} Y_s] \end{aligned}$$

$$= Z_x + s [Z_{xx}(t) + Z_{xy}(2)] + 3 [Z_{yx}(t) + Z_{yy}(2)]$$

$$= Z_x + st Z_{xx} + 2s Z_{xy} + 3t Z_{yx} + 6 Z_{yy}$$

$$= Z_x + st Z_{xx} + (2s + 3t) Z_{xy} + 6 Z_{yy}$$



Find all points on the surface $x^2 - y^2 - z^2 = 4$ where the tangent plane is parallel to the plane $3x + 2y + z = 7$

Oct 2018
Question 10

$$f(x, y, z) = x^2 - y^2 - z^2 - 4$$

$$f_x = 2x$$

$$f_y = -2y$$

$$f_z = -2z$$

$$\implies \nabla f = \langle 2x, -2y, -2z \rangle$$

The normal line to the plane $3x + 2y + z = 7$

$$n = \langle 3, 2, 1 \rangle$$

$$\because \nabla f \parallel n$$

$$\therefore \langle 2x, -2y, -2z \rangle = k \langle 3, 2, 1 \rangle$$

$$2x = 3k \implies x = \frac{3}{2}k$$

$$-2y = 2k \implies y = -k$$

$$-2z = k \implies z = -\frac{1}{2}k$$

Sub on S

$$\left(\frac{3}{2}k\right)^2 - (-k)^2 - \left(-\frac{1}{2}k\right)^2 = 4$$

$$\frac{9}{4}k^2 - k^2 - \frac{1}{4}k^2 = 4$$

$$\frac{9}{4}k^2 - \frac{4}{4}k^2 - \frac{1}{4}k^2 = 4$$

$$\frac{4}{4}k^2 = 4$$

$$k^2 = 4$$

$$k = -2$$

$$k = 2$$

$$k = -2$$

$$x = \frac{3}{2}(-2) = -3$$

$$y = -(-2) = 2$$

$$z = -\frac{1}{2}(-2) = 1$$

$$k = 2$$

$$x = 3$$

$$y = -2$$

$$z = -1$$

$$\implies P(-3, 2, 1)$$

$$\implies P(3, -2, -1)$$

Math 211 First Exam (Version A: Sections 1, 3, 51, and 53)

23 June 2018

Questions 1–5: 5 points each, questions 6–10: 15 points each.

1. The unit tangent vector to the curve $\mathbf{r}(t) = (t^2 - 4t)\mathbf{i} + (t^3 - t + 2)\mathbf{j} + (t^3 + 1)\mathbf{k}$ at the point $(5, 2, 0)$ is

(a) $-\frac{6}{7}\mathbf{i} + \frac{2}{7}\mathbf{j} + \frac{3}{7}\mathbf{k}$

(c) $\frac{6}{7}\mathbf{i} - \frac{2}{7}\mathbf{j} - \frac{3}{7}\mathbf{k}$

(b) $\frac{5}{\sqrt{29}}\mathbf{i} + \frac{2}{\sqrt{29}}\mathbf{j}$

(d) $-\frac{2}{\sqrt{17}}\mathbf{i} + \frac{2}{\sqrt{17}}\mathbf{j} + \frac{3}{\sqrt{17}}\mathbf{k}$

2. The arc-length of the curve $\mathbf{r}(t) = -4\cos t\mathbf{i} + 2t\mathbf{j} + 4\sin t\mathbf{k}$ between the points $(-4, 0, 0)$ and $(4, 2\pi, 0)$ is

(a) π

(b) $\pi/2$

(c) $\sqrt{5}\pi$

(d) $2\sqrt{5}\pi$

3. The curvature at any point of the curve $\mathbf{r}(t) = 2\cos t\mathbf{i} + 2\sin t\mathbf{j} + 10\mathbf{k}$ is

(a) 2

(b) 3

(c) $1/2$

(d) $1/3$

4. The limit of $f(x, y) = \frac{xy^2}{x^2 + y^4}$ as $(x, y) \rightarrow (0, 0)$ is

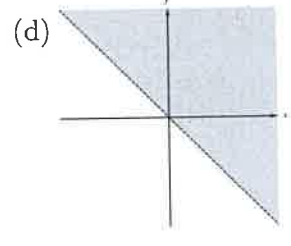
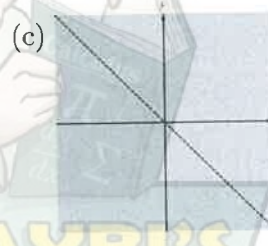
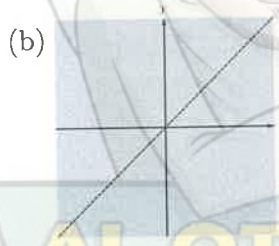
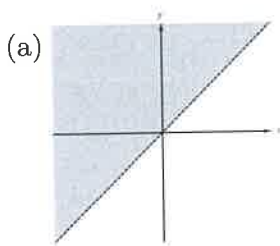
(a) 0

(b) $1/2$

(c) 1

(d) Undefined

5. Which of the following regions is the domain of $f(x, y) = \frac{\cos(x+y)}{x-y}$?



6. Find the value and direction of the maximum directional derivative of $f(x, y) = xy^2 - 2\sqrt{x}$ at the point $(1, 2)$.

7. Find an equation for the tangent plane to the surface $x + y + z = e^{xyz}$ at the point $(3, 0, -2)$.

8. Let $z = f(x, y)$ where $x = s + t$, $y = s - t$, and f is a differentiable function. Show that

$$\left(\frac{\partial f}{\partial x}\right)^2 - \left(\frac{\partial f}{\partial y}\right)^2 = \frac{\partial z}{\partial s} \cdot \frac{\partial z}{\partial t}.$$

9. Find and classify all critical points of the function $f(x, y) = 4 + x^3 + y^3 - 3xy$.

10. Find the maximum value of $f(x, y, z) = xyz$ on the part of the paraboloid $x^2 + y^2 + z = 16$ that lies in the first octant.

The limit of $f(x,y) = \frac{xy^2}{x^2 + y^4}$ as $(x,y) \rightarrow (0,0)$ is

- (a) 0 (b) $\frac{1}{2}$ (c) 1 (d) Undefined

June 2018
Question #4
Version A

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4} \quad \frac{0}{0}$$

along $x=0$ (y-axis)

$$\lim_{y \rightarrow 0} \frac{0}{0 + y^4} = \boxed{0} = L_1$$

along $x = y^2$

$$\lim_{y \rightarrow 0} \frac{y^2 y^2}{(y^2)^2 + y^4}$$

$$\lim_{y \rightarrow 0} \frac{y^4}{2y^4} = \boxed{\frac{1}{2}} = L_2$$

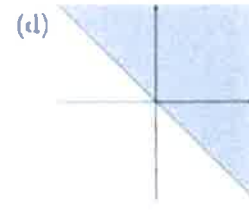
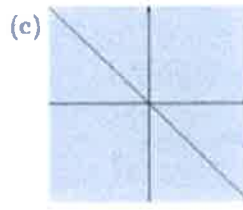
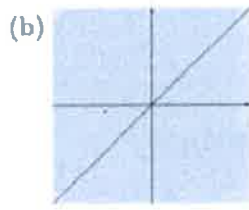
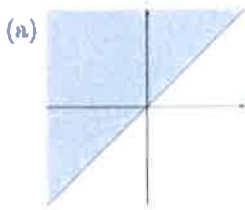
$$\therefore L_1 \neq L_2$$

$$\therefore \lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4} \text{ DNE (two path rule)}$$

(d) ✓

5. Which of the following regions is the domain of $f(x, y) = \frac{\cos(x+y)}{x-y}$?

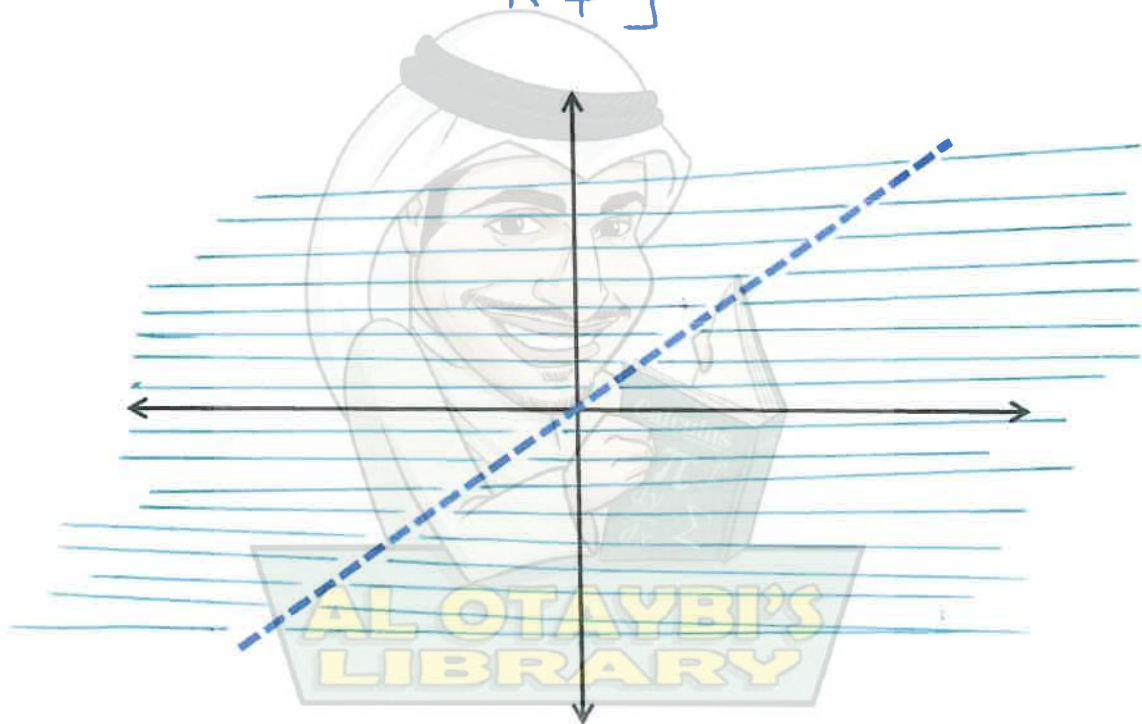
June 2018
Version A
Question #5



$$f(x, y) = \frac{\cos(x+y)}{x-y}$$

$$D_f = \{(x, y) \in \mathbb{R}^2 : x - y \neq 0\}$$

$$x \neq y$$



(b)

Find the value and direction of the maximum directional derivative of $f(x, y) = xy^2 - 2\sqrt{x}$ at the point $(1, 2)$

June 2018
Version A
Question 6

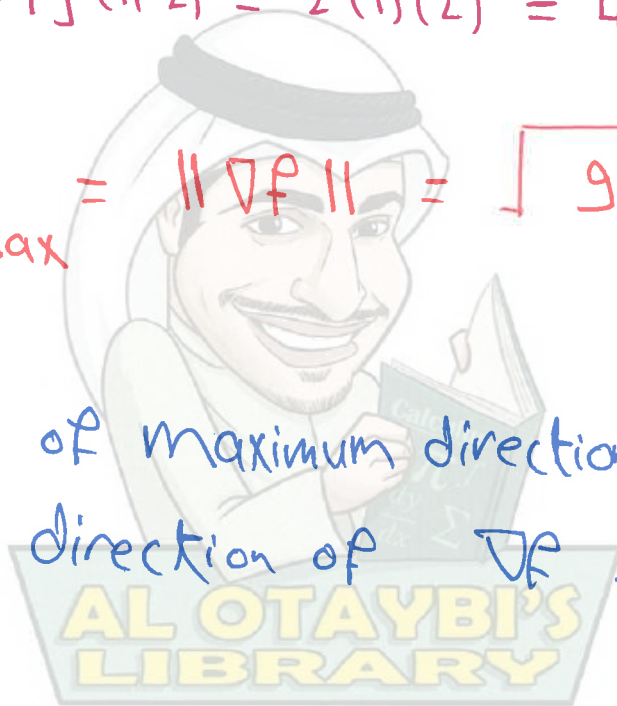
$$f(x, y) = xy^2 - 2\sqrt{x} \quad P(1, 2)$$

$$f_x = y^2 - \frac{1}{\sqrt{x}} \longrightarrow f_x(1, 2) = 4 - 1 = 3$$

$$f_y = 2xy \longrightarrow f_y(1, 2) = 2(1)(2) = 4 \implies \nabla f = \langle 3, 4 \rangle$$

$$[D_u f(1, 2)]_{\max} = \|\nabla f\| = \sqrt{9 + 16} = \sqrt{25} = 5$$

The direction of maximum directional derivative is the same direction of $\nabla f = \langle 3, 4 \rangle$



Find an equation for the $\downarrow$ tangent plane to the surface $x + y + z = e^{xyz}$
at the point $(3, 0, -2)$

June 2018
Version A
Question 7

$$f(x, y, z) = x + y + z - e^{xyz} \quad P(3, 0, -2)$$

$$f_x = 1 - yze^{xyz} \longrightarrow f_x(3, 0, -2) = 1 - 0 = 1$$

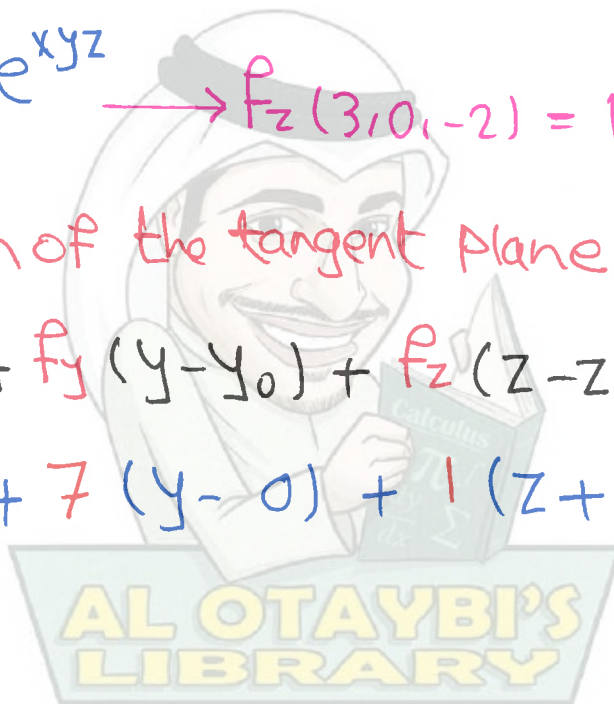
$$f_y = 1 - xze^{xyz} \longrightarrow f_y(3, 0, -2) = 1 - 3(-2)e^0 = 7$$

$$f_z = 1 - xye^{xyz} \longrightarrow f_z(3, 0, -2) = 1 - 0 = 1$$

$\therefore$ The equation of the tangent plane

$$f_x(x - x_0) + f_y(y - y_0) + f_z(z - z_0) = 0$$

$$1(x - 3) + 7(y - 0) + 1(z + 2) = 0$$



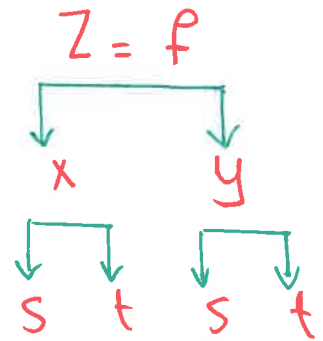
Let $z = f(x, y)$ where $x = s + t$, $y = s - t$ and f is a differentiable function, Show that

$$\left(\frac{\partial f}{\partial x}\right)^2 - \left(\frac{\partial f}{\partial y}\right)^2 = \frac{\partial z}{\partial s} \cdot \frac{\partial z}{\partial t}$$

June 2018
Question #8

$$(f_x)^2 - (f_y)^2 = z_s \cdot z_t \quad ??$$

$$\begin{aligned} z_s &= z_x x_s + z_y y_s \\ &= z_x (1) + z_y (1) \end{aligned}$$



$$\begin{aligned} z_t &= z_x x_t + z_y y_t \\ &= z_x (1) + z_y (-1) \end{aligned}$$

$$\begin{aligned} \text{R.H.S} &= z_s \cdot z_t \\ &= (z_x + z_y) \cdot (z_x - z_y) \\ &= (z_x)^2 - \cancel{z_x z_y} + \cancel{z_y z_x} - (z_y)^2 \\ &= (z_x)^2 - (z_y)^2 \\ &= (f_x)^2 - (f_y)^2 = \text{L.H.S} \end{aligned}$$

Find and classify all critical points of the function

$$f(x, y) = 4 + x^3 + y^3 - 3xy$$

June 2018

Version A

Question 9

$$f_x = 3x^2 - 3y \begin{cases} f_{xx} = 6x \\ f_{xy} = -3 \end{cases}$$

$$f_y = 3y^2 - 3x \longrightarrow f_{yy} = 6y$$

$$f_x = 0$$

$$3x^2 - 3y = 0 \longrightarrow 3y = 3x^2 \longrightarrow y = x^2 \quad (1)$$

$$f_y = 0$$

$$3y^2 - 3x = 0 \xrightarrow{\div 3} y^2 - x = 0$$

(1) in (2)

$$(x^2)^2 - x = 0 \longrightarrow x^4 - x = 0$$

$$x(x^3 - 1) = 0 \begin{cases} x = 0 \longrightarrow y = 0 \\ x = 1 \longrightarrow y = 1 \end{cases} \begin{matrix} P(0,0) \\ P(1,1) \end{matrix}$$

∴ The critical points are (0,0) and (1,1)

$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx} f_{yy} - (f_{xy})^2$$

$$= 6x(6y) - (-3)^2 = 36xy - 9 = 9[4xy - 1]$$

critical points	$D = 9[4xy - 1]$	$f_{xx} = 6x$	result
(0,0)	$-9 < 0$	—	Saddle point
(1,1)	$3 > 0$	$6 > 0$	Local minimum No L. max

Find the maximum value of $f(x, y, z) = xyz$ on the part of paraboloid $x^2 + y^2 + z = 16$ that lies in the first octant

June 2018
Version A
Question 10

$$f(x, y, z) = xyz$$

$$g(x, y, z) = x^2 + y^2 + z - 16 = 0$$

$$\nabla f = \langle yz, xz, xy \rangle$$

$$\nabla g = \langle 2x, 2y, 1 \rangle$$

$$\rightarrow \nabla f = \lambda \nabla g$$

$$\langle yz, xz, xy \rangle = \lambda \langle 2x, 2y, 1 \rangle$$

$$\rightarrow yz = 2x\lambda$$

(1)

$$xz = 2y\lambda$$

(2)

$$xy = \lambda$$

(3)

(3) in (1)

$$yz = 2x(xy)$$

$$\rightarrow yz = 2x^2y$$

$$2x^2y - yz = 0$$

$$y[2x^2 - z] = 0$$

$$\rightarrow y = 0$$

∉ first octant.

$$\rightarrow z = 2x^2$$

$$\rightarrow x^2 = \frac{1}{2}z$$

$$x = -\frac{1}{\sqrt{2}}\sqrt{z}$$

∉ first octant

$$x = \frac{1}{\sqrt{2}}\sqrt{z}$$

(3) in (2)

$$xz = 2y(xy)$$

$$\rightarrow xz = 2xy^2$$

$$2xy^2 - xz = 0$$

$$x[2y^2 - z] = 0$$

$$\rightarrow x = 0$$

∉ first octant

$$\rightarrow z = 2y^2$$

$$y^2 = \frac{1}{2}z$$

$$y = -\frac{1}{\sqrt{2}}\sqrt{z}$$

∉ first octant

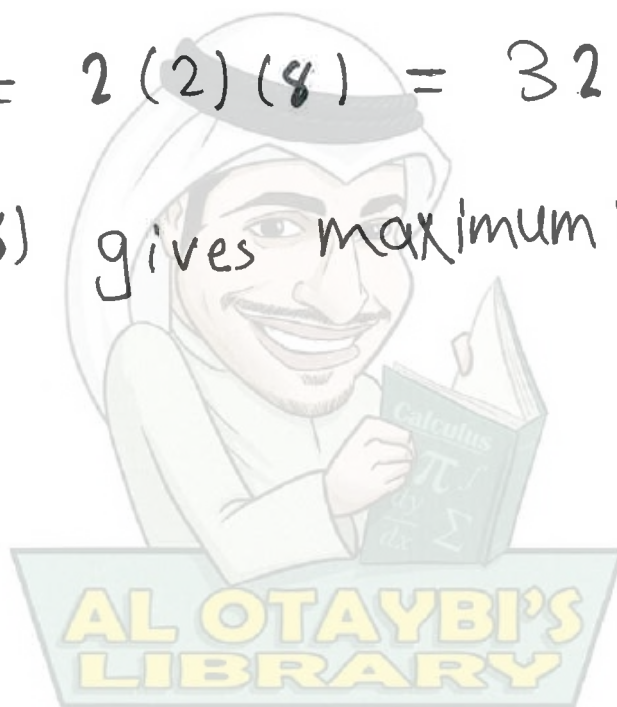
$$y = \frac{1}{\sqrt{2}}\sqrt{z}$$

$$\begin{aligned}
 x &= \frac{1}{\sqrt{2}} \sqrt{z} \\
 y &= \frac{1}{\sqrt{2}} \sqrt{z}
 \end{aligned}
 \xrightarrow{\text{in } g}
 \begin{aligned}
 \frac{1}{2}z + \frac{1}{2}z + z - 16 &= 0 \\
 z + z &= 16 \\
 2z &= 16 \rightarrow z = 8
 \end{aligned}$$

$$\begin{aligned}
 z = 8 & \begin{cases} \rightarrow x = \frac{1}{\sqrt{2}} \sqrt{8} = \sqrt{\frac{8}{2}} = \sqrt{4} = 2 \\ \rightarrow y = \frac{1}{\sqrt{2}} \sqrt{8} = \sqrt{\frac{8}{2}} = \sqrt{4} = 2 \end{cases} \xrightarrow{\quad} P(2, 2, 8)
 \end{aligned}$$

$$f(2, 2, 8) = 2(2)(8) = 32 \quad \text{max value}$$

$\therefore P(2, 2, 8)$ gives maximum value of f .



Math 211 First Exam (Version B: Sections 2, 52, and 54)

23 June 2018

Questions 1-5: 5 points each, questions 6-10: 15 points each.

1. The unit tangent vector to the curve $\mathbf{r}(t) = -(t^2 + 4t)\mathbf{i} + (t^3 + 3t)\mathbf{j} + (t^3 + 1)\mathbf{k}$ at the point $(3, 2, 0)$ is

(a) $\frac{2}{7}\mathbf{i} - \frac{6}{7}\mathbf{j} - \frac{3}{7}\mathbf{k}$

(c) $-\frac{2}{7}\mathbf{i} + \frac{6}{7}\mathbf{j} + \frac{3}{7}\mathbf{k}$

(b) $\frac{3}{\sqrt{13}}\mathbf{i} + \frac{2}{\sqrt{13}}\mathbf{j}$

(d) $-\frac{2}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} + \frac{1}{3}\mathbf{k}$

2. The arc-length of the curve $\mathbf{r}(t) = -4\cos t\mathbf{i} + 2t\mathbf{j} + 4\sin t\mathbf{k}$ between the points $(-4, 0, 0)$ and $(0, \pi, 4)$ is

(a) π

(b) $\pi/2$

(c) $\sqrt{5}\pi$

(d) $2\sqrt{5}\pi$

3. The curvature at any point of the curve $\mathbf{r}(t) = 3\cos t\mathbf{i} + 3\sin t\mathbf{j} + 10\mathbf{k}$ is

(a) 2

(b) 3

(c) $1/2$

(d) $1/3$

4. The limit of $f(x, y) = \frac{xy^2}{x^2 + y^4}$ as $(x, y) \rightarrow (0, 0)$ is

(a) 0

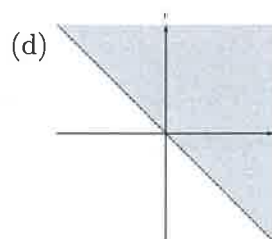
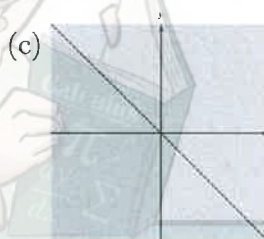
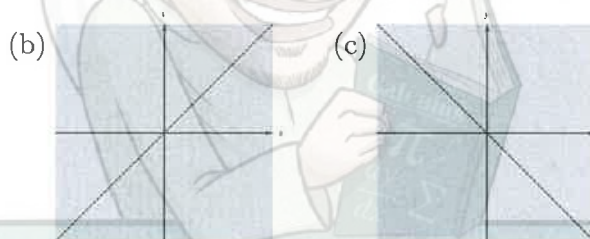
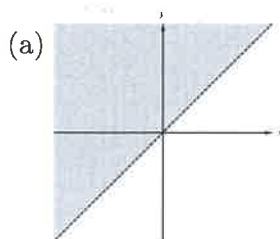
(b) $1/2$

(c) 1

(d) Undefined

تم حله بالامتحان
السابق

5. Which of the following regions is the domain of $f(x, y) = (y - x)\ln(y + x)$?



6. Find the value and direction of the maximum directional derivative of $f(x, y) = xy^2 - 2\sqrt{x}$ at the point $(1, 1)$.

تم حله بالامتحان السابق (الاختلاف فقط بالتقارر)

7. Find an equation for the tangent plane to the surface $x + y + z = \ln(1 + xyz)$ at the point $(0, -2, 2)$.

8. Let $w = f(u, v)$, where $u = xy$, $v = \frac{x}{y}$, and f is a differentiable function. Show that

$$x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} = 2xy \frac{\partial f}{\partial u}$$

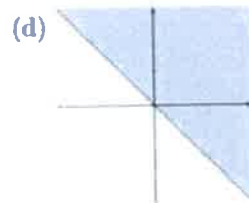
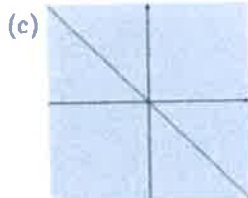
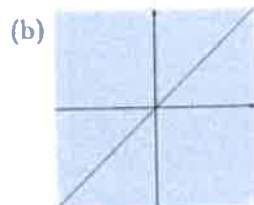
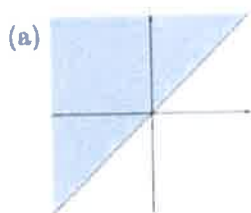
9. Find and classify all critical points of the function $f(x, y) = 6 - x^4 + 2x^2 - y^2$.

10. Find the maximum value of $f(x, y, z) = xyz$ on the part of the paraboloid $x^2 + y^2 + z = 1$ that lies in the first octant.

تم حله بالامتحان السابق
(الاختلاف فقط بالهرف الايمن للمعادلة المساعدة)

5. Which of the following regions is the domain of $f(x, y) = (y - x) \ln(y + x)$?

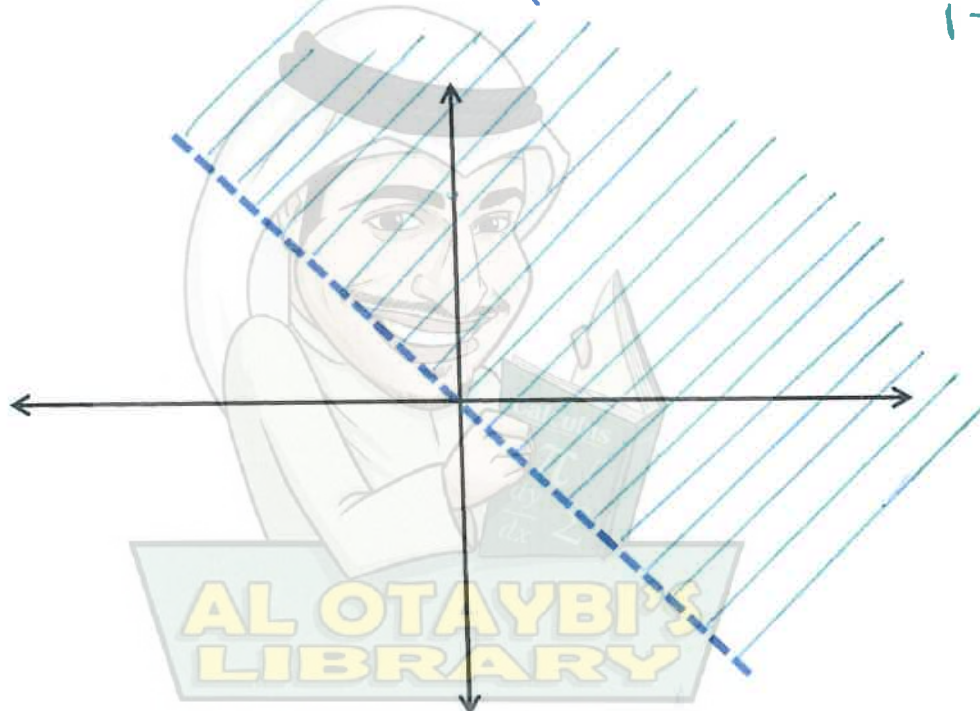
June 2018
Version B
Question #5



$$f(x, y) = (y - x) \ln(y + x)$$

$$D_f = \{(x, y) \in \mathbb{R}^2 : y + x > 0, y > -x\}$$

$$(0, 1) \\ 1 > 0 \\ \checkmark$$



(d)

Find an equation for the tangent plane to the surface

$x + y + z = \ln(1 + xyz)$ at the point $(0, -2, 2)$

June 2018

Version B

Question 7

$$f(x, y, z) = x + y + z - \ln(1 + xyz) \quad P(0, -2, 2)$$

$$f_x = 1 - \frac{yz}{1 + xyz} \rightarrow f_x(0, -2, 2) = 1 - \frac{-4}{1+0} = 5$$

$$f_y = 1 - \frac{xz}{1 + xyz} \rightarrow f_y(0, -2, 2) = 1 - 0 = 1$$

$$f_z = 1 - \frac{xy}{1 + xyz} \rightarrow f_z(0, -2, 2) = 1 - 0 = 1$$

$\therefore$ The equation of the tangent plane

$$f_x(x - x_0) + f_y(y - y_0) + f_z(z - z_0) = 0$$

$$5(x - 0) + (1)(y + 2) + (1)(z - 2) = 0$$

Let $w = f(u, v)$ where $u = xy$, $v = \frac{x}{y}$ and f is a differentiable function, Show that

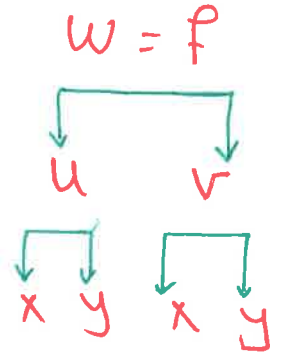
$$x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} = 2xy \frac{\partial f}{\partial u}$$

June 2018
Question #8

$$x w_x + y w_y = 2xy f_u \quad ?$$

$$\begin{aligned} w_x &= w_u u_x + w_v v_x \\ &= w_u y + w_v \frac{1}{y} \end{aligned}$$

$$\begin{aligned} w_y &= w_u u_y + w_v v_y \\ &= w_u x + w_v \left(-\frac{x}{y^2}\right) \end{aligned}$$



$$\begin{aligned} \text{R.H.S} &= x w_x + y w_y \\ &= x \left[w_u y + w_v \frac{1}{y} \right] + y \left[w_u x + w_v \left(-\frac{x}{y^2}\right) \right] \\ &= xy w_u + \frac{x}{y} w_v + xy w_u - \frac{x}{y} w_v \\ &= 2xy w_u \\ &= 2xy f_u = \text{R.H.S} \end{aligned}$$

Find and classify all critical points of the function

June 2018

Version B

Question 9

$$f(x, y) = 6 - x^4 + 2x^2 - y^2$$

$$f_x = -4x^3 + 4x \begin{cases} f_{xx} = -12x^2 + 4 \\ f_{xy} = 0 \end{cases}$$

$$f_y = -2y \longrightarrow f_{yy} = -2$$

$$f_x = 0$$

$$-4x^3 + 4x = 0 \xrightarrow{\div 4} -x^3 + x = 0$$

$$x[-x^2 + 1] = 0 \begin{cases} x = 0 \\ x^2 = 1 \end{cases} \begin{cases} x = -1 \\ x = 1 \end{cases}$$

$$f_y = 0$$

$$-2y = 0 \longrightarrow y = 0$$

∴ The critical points are (0,0) (-1,0) and (1,0)

$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx}f_{yy} - (f_{xy})^2$$

$$= (-12x^2 + 4)(-2) - 0$$

$$= 4(-3x^2 + 1)(-2) = -8(-3x^2 + 1)$$

Critical Points	$D = -8(-3x^2 + 1)$	$f_{xx} = -12x^2 + 4$	result
(0,0)	$-8 < 0$	—	Saddle point
(-1,0)	$16 > 0$	$-8 < 0$	Local maximum
(1,0)	$16 > 0$	$-8 < 0$	Local maximum

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No Local Minimum